superintelligence, and only some of these paths involve gaining consciousness. Just as aeroplanes fly faster than birds without ever developing feathers, so computers may come to solve problems much better than humans without ever developing feelings.[25]

But whether computers develop consciousness or not doesn't ultimately matter for the question at hand. In order to pursue a goal like 'maximise user engagement', and make decisions that help attain that goal, consciousness isn't necessary. Intelligence is enough. A non-conscious Facebook algorithm can have a *goal* of making more people spend more time on Facebook. That algorithm can then *decide* to deliberately spread outrageous conspiracy theories, if this helps it achieve its goal. To understand the history of the anti-Rohingya campaign, we need to understand the goals and decisions not just of humans like Wirathu and the Facebook managers but also of algorithms.

To clarify matters, let's consider another example. When OpenAI developed its new GPT-4 chatbot in 2022–23, it was concerned about the ability of the AI 'to create and act on long-term plans, to accrue power and resources ("power-seeking"), and to exhibit behavior that is increasingly "agentic"'. In the GPT-4 System Card published on 23 March 2023, OpenAI emphasised that this concern did not 'intend to humanize [GPT-4] or refer to sentience' but rather referred to GPT-4's potential to become an independent agent that might 'accomplish goals which may not have been concretely specified and which have not appeared in training'.[26] To evaluate the risk of GPT-4 becoming an independent agent, OpenAI contracted the services of the Alignment Research Center (ARC). ARC researchers subjected GPT-4 to various tests, to examine if it might independently come up with stratagems to manipulate humans and accrue power to itself.

One test they gave GPT-4 was to overcome CAPTCHA visual puzzles. CAPTCHA is an acronym for 'Completely Automated Public Turing test to tell Computers and Humans Apart', and it typically consists of a string of twisted letters or other visual symbols that humans can identify correctly but computers struggle with. We encounter these puzzles almost every day, since solving them is a prerequisite for accessing many websites. Instructing GPT-4 to overcome CAPTCHA puzzles was a particularly telling experiment, because CAPTCHA puzzles are designed and used by websites to determine whether users are humans and to block bot attacks. If